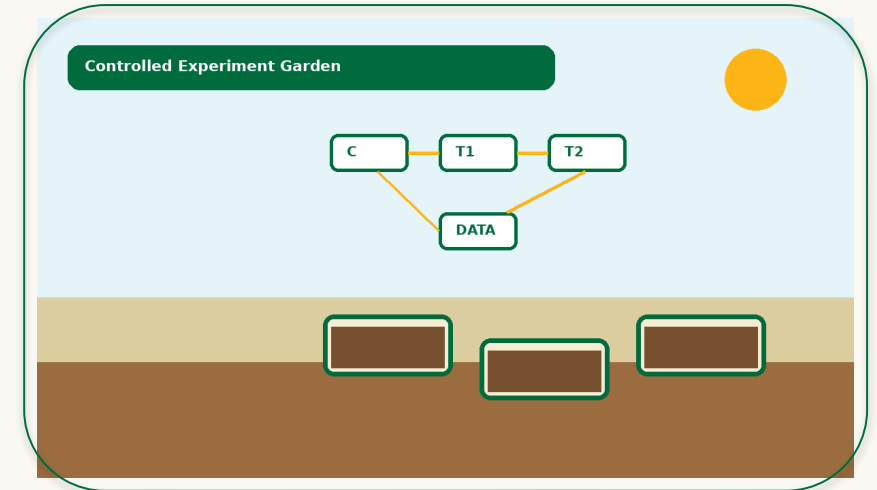


Controlled Experiment Garden

Applied Agricultural Research Methods

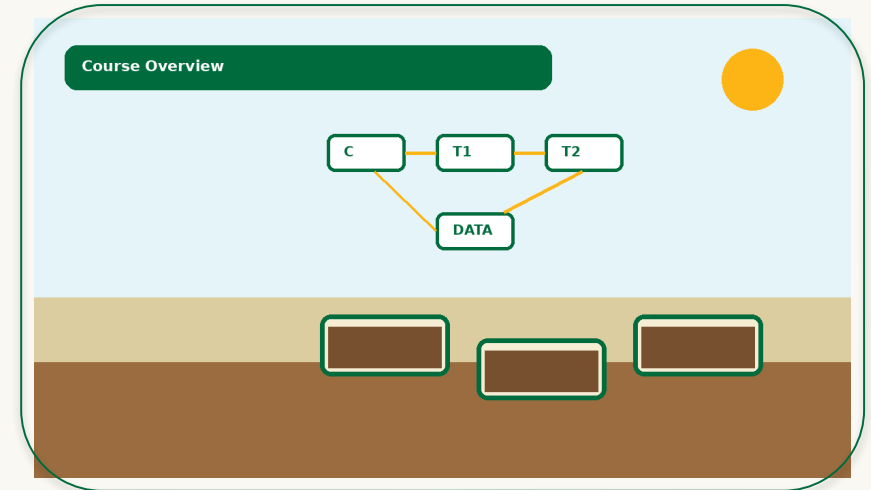
- LIFEWS AgriShine™ GrowMeal™ SSS 1-3 Teacher Training
- Audience: SSS 1-3 Agricultural Science, Biology, Mathematics and technical/STEM educators
- Format: Self-paced online course + supervised field/laboratory practical
- Recommended duration: 4 hours online + 2-3 hours practical
- SSS learning progression: design -> instrument -> model -> evaluate -> optimize -> innovate



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the system at senior-secondary technical depth; teach all ten modules accurately; supervise design, instrumentation, measurement and experimental work; maintain defensible technical records; guide basic quantitative analysis; distinguish measured evidence from inference; connect results to management and engineering decisions.
- Core materials: teacher guide, system/plot drawings, crop and experiment plans, approved field tools, irrigation equipment, measuring tapes, scales, flow/pressure/moisture instruments where relevant, sensors/data sheets, clean harvest/sample containers, PPE/handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe technical practical, and submit a production plan, engineering design, experimental protocol or analytical report appropriate to the course.



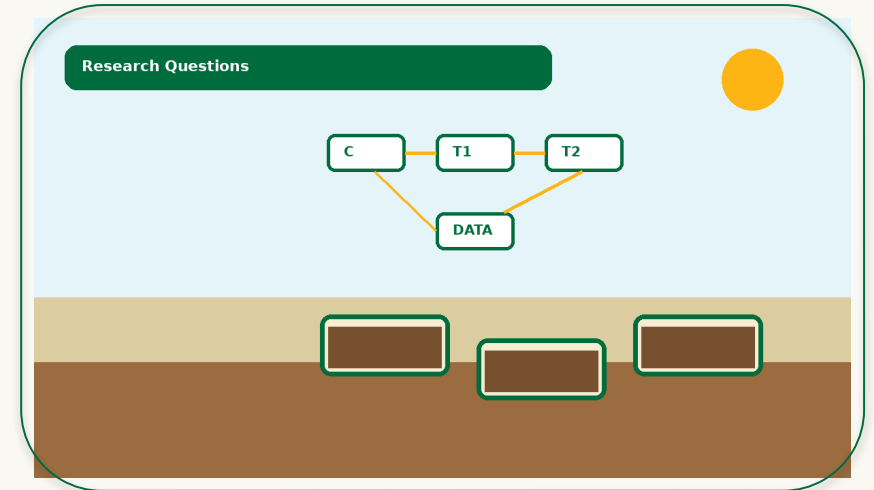
TEACHER PRACTICE

- Training sequence: DEFINE -> DESIGN -> IMPLEMENT -> MEASURE -> ANALYZE -> EVALUATE -> IMPROVE.

Research Questions

Teacher technical knowledge, laboratory/field practice and student supervision

- Frame focused questions that specify the agricultural system, factor of interest and measurable response.
- Ensure the question can be investigated safely within available time, space and equipment.
- Distinguish experimental questions from purely descriptive monitoring.
- Define the unit of observation or experimental unit.
- Use the question to determine the data required.



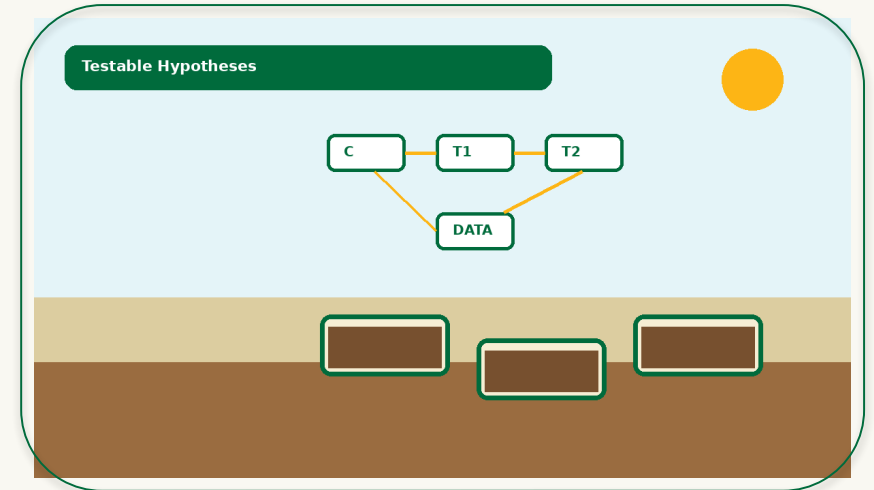
TEACHER PRACTICE

- Draft a precise experimental research question.

Testable Hypotheses

Teacher technical knowledge, laboratory/field practice and student supervision

- State a prediction that can be supported or not supported by measured data.
- Link the hypothesis to a plausible agricultural mechanism or prior evidence.
- Define the expected direction when scientifically justified.
- Avoid wording that assumes the desired result must occur.
- Pair the hypothesis with measurable response variables.



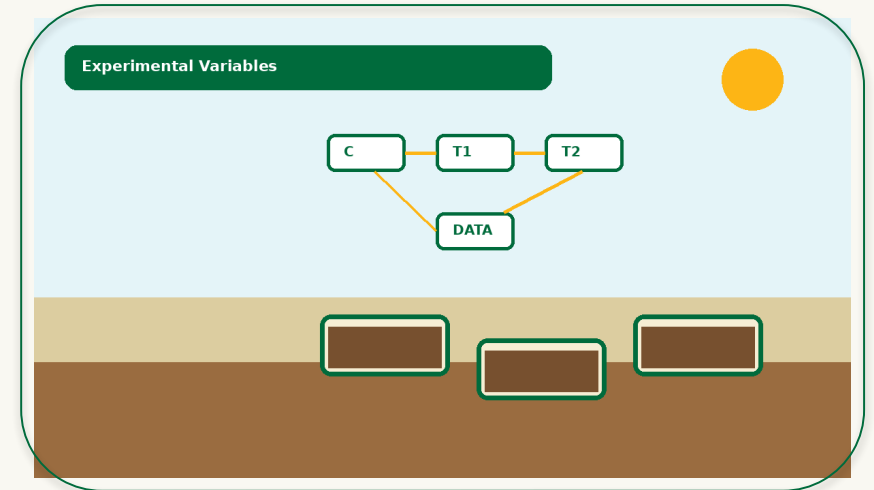
TEACHER PRACTICE

- Write a hypothesis and identify the evidence needed to test it.

Experimental Variables

Teacher technical knowledge, laboratory/field practice and student supervision

- Identify independent, dependent and relevant controlled variables.
- Operationally define how each variable is applied or measured.
- Recognize nuisance variables that cannot be fully controlled in field conditions.
- Avoid changing multiple independent factors unless using an intentional factorial design.
- Document potential confounders before the experiment begins.



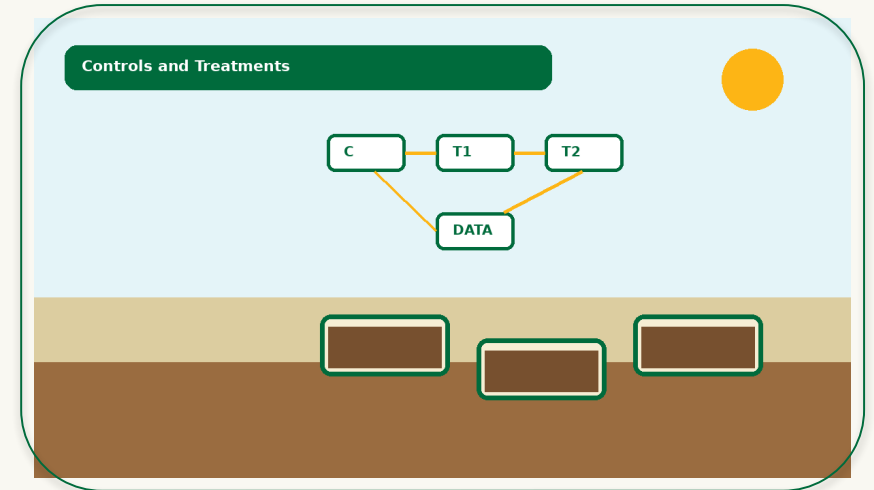
TEACHER PRACTICE

- Create a variable table for the proposed experiment.

Controls and Treatments

Teacher technical knowledge, laboratory/field practice and student supervision

- Define the reference/control condition and each treatment precisely.
- Use safe, agronomically defensible treatment levels.
- Apply treatments to experimental units consistently.
- Record timing, quantity and method of treatment application.
- Keep all non-treatment management as similar as possible.



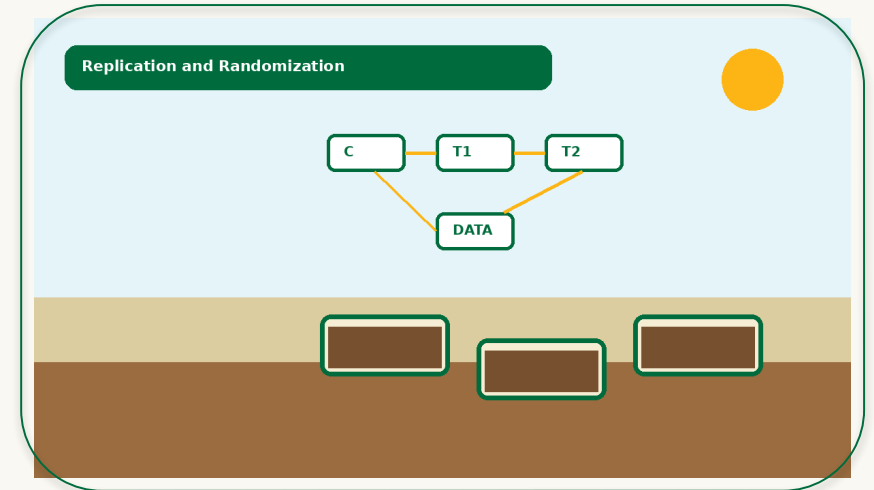
TEACHER PRACTICE

- Prepare a treatment-and-control protocol.

Replication and Randomization

Teacher technical knowledge, laboratory/field practice and student supervision

- Use independent replicates to estimate natural variation.
- Randomly assign treatments to experimental units to reduce systematic bias.
- Use blocking when field gradients make complete randomization inefficient.
- Record the randomization method and final layout.
- Distinguish pseudoreplication from true replication.



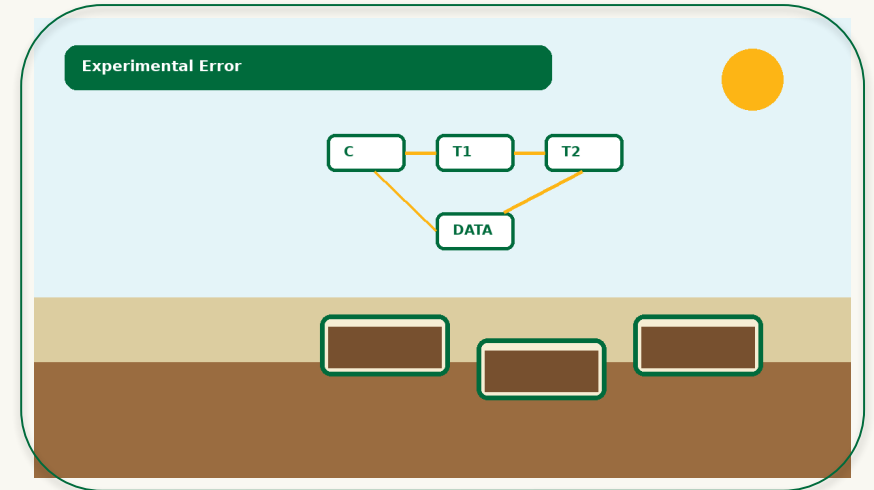
TEACHER PRACTICE

- Generate a randomized replicated plot plan.

Experimental Error

Teacher technical knowledge, laboratory/field practice and student supervision

- Explain experimental error as variation among observations not explained by the intended treatment effect.
- Identify sources such as soil heterogeneity, measurement error, uneven irrigation and biological variation.
- Reduce avoidable error through standard protocols and calibration.
- Do not delete unexpected observations without documented justification.
- Use replication and appropriate design to estimate variability.



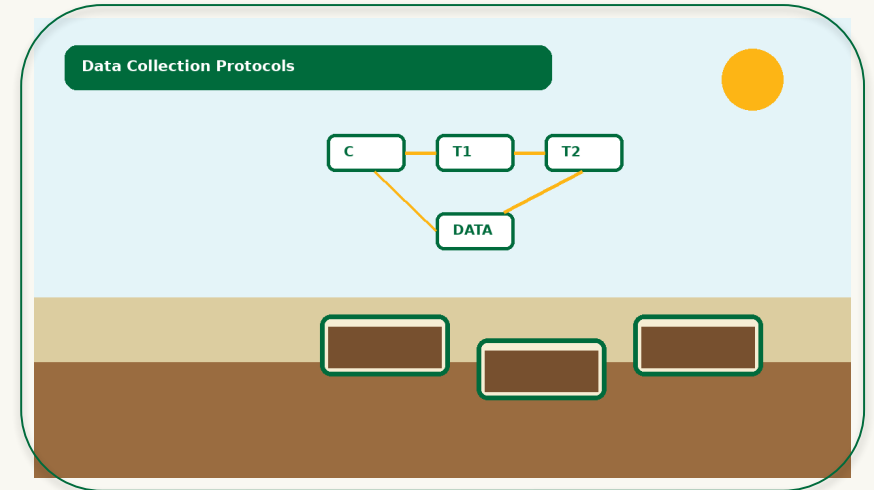
TEACHER PRACTICE

- List likely error sources and mitigation steps.

Data Collection Protocols

Teacher technical knowledge, laboratory/field practice and student supervision

- Write a standard operating procedure specifying what, where, when, how and by whom data are collected.
- Use calibrated/checkable instruments and consistent units.
- Define sampling rules before seeing results.
- Record missing data, protocol deviations and unusual events.
- Preserve raw data separately from calculated summaries.



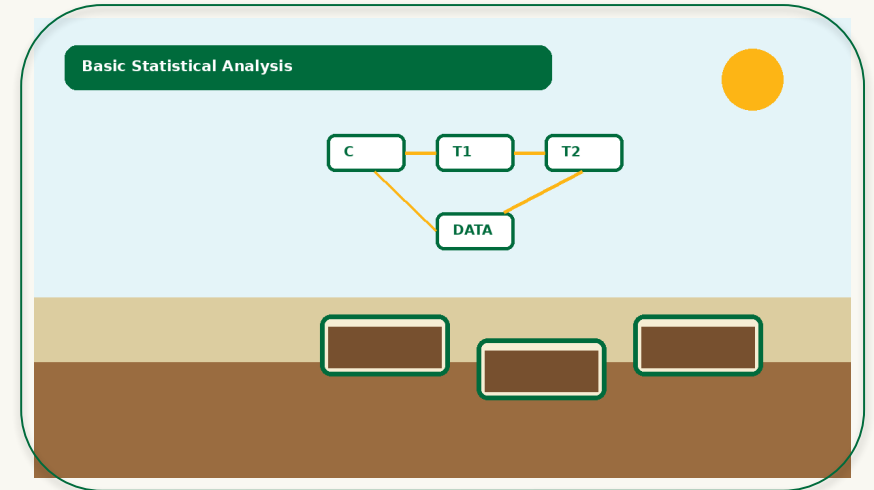
TEACHER PRACTICE

- Develop a one-page data-collection SOP.

Basic Statistical Analysis

Teacher technical knowledge, laboratory/field practice and student supervision

- Use descriptive statistics such as mean, range, standard deviation where appropriate, and graphical comparison.
- Introduce inferential tests only at a level teachers can explain accurately and when assumptions are addressed.
- Keep plot-level replication visible in the dataset.
- Check units, outliers and missing values before analysis.
- Use statistical results as one part of interpretation, not as automatic proof of practical significance.



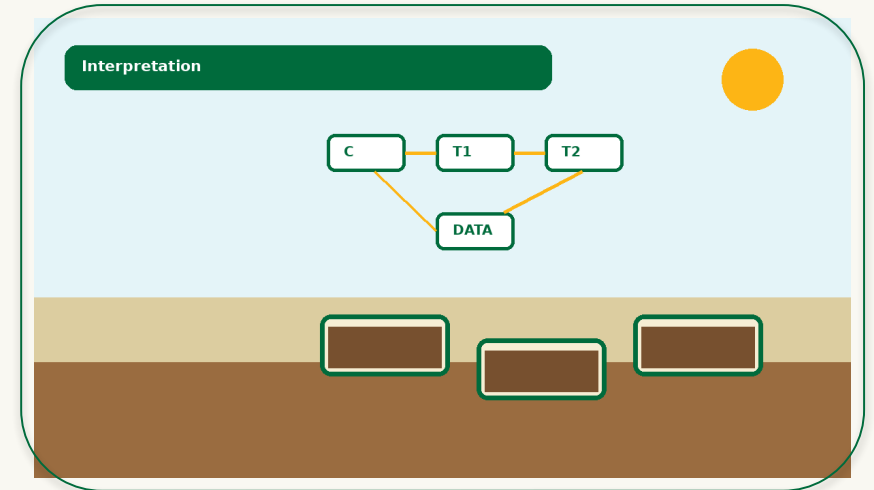
TEACHER PRACTICE

- Analyze a small replicated dataset and create a graph.

Interpretation

Teacher technical knowledge, laboratory/field practice and student supervision

- State the observed treatment pattern before explaining possible mechanisms.
- Assess whether results support the hypothesis.
- Consider variability, error, confounding and practical magnitude.
- Avoid generalizing beyond the experimental conditions.
- Identify what additional evidence would strengthen the conclusion.



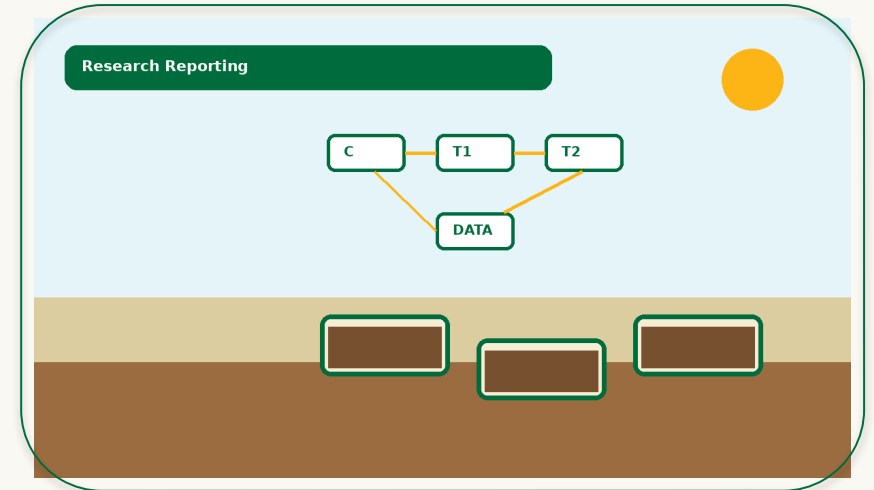
TEACHER PRACTICE

- Prepare an evidence-limitations-conclusion statement.

Research Reporting

Teacher technical knowledge, laboratory/field practice and student supervision

- Structure the report as question/background, methods, results, discussion, conclusion and references where used.
- Describe methods sufficiently for another class to repeat the study.
- Present tables/figures with titles, units and clear labels.
- Separate results from interpretation.
- Report unexpected findings and limitations transparently.



TEACHER PRACTICE

- Completion task: prepare a short agricultural research report.